HW 5 — Jackson 8.2 & 8.4

black = write it down · **red = the answer** · *teal = where it came from / my note* · grey = optional

What the lectures give you, and what they don't. Lec 10 sets up all three mode families — TEM, TE, TM — and slide 16 even shows the coaxial line. But the quantities these two problems ask for (power, attenuation, impedance, R and L, Bessel cutoffs) are **not** in any lecture. Every such step below is tagged Jackson §8.1 so you can see which parts are lecture-backed.

Lec 10 s13–16	TEM mode: $\nabla_t\cdot\vec{E}_t = 0$, $\nabla_t\times\vec{E}_t = 0$, $\vec{E}_t = -\nabla\phi$, $\nabla_t^2\phi = 0$, $k = \omega\sqrt{\mu\varepsilon}$	**8.2**
Lec 10 s17–18	TM: $\nabla_t^2 E_z + \gamma^2 E_z = 0$, $E_z = 0$ on S | TE: $\nabla_t^2 B_z + \gamma^2 B_z = 0$, $\partial B_z/\partial n = 0$ on S	**8.4**
not lectured	*Poynting power flow, wall loss and attenuation, characteristic impedance, R and L per unit length, Bessel functions*	**both**

Everything below was checked numerically: each 8.2 result was recomputed by direct integration of the fields (agreement to 16 digits), and the 8.4 attenuation formulas were checked against a direct evaluation of the wall-loss and power integrals (also 16 digits).

Jackson 8.2 — Coaxial transmission line

A transmission line of two concentric metal cylinders (conductivity σ, skin depth δ), inner radius a, outer radius b, filled with a lossless dielectric (μ, ε), carrying a **TEM mode**. H_0 is the peak azimuthal magnetic field at the surface of the inner conductor. Find **(a)** the time-averaged power flow, **(b)** the attenuation, **(c)** the characteristic impedance, **(d)** the series resistance and inductance per unit length.

ANSWER

$$\textbf{(a)}\quad P = \sqrt{\mu/\varepsilon}\,\pi a^2 |H_0|^2 \ln(b/a)$$

$$\textbf{(b)}\quad P(z) = P_0 e^{-2\gamma z},\quad \gamma = \frac{1}{2\sigma\delta}\sqrt{\varepsilon/\mu}\,\frac{(1/a + 1/b)}{\ln(b/a)}$$

$$\textbf{(c)}\quad Z_0 = \frac{1}{2\pi}\sqrt{\mu/\varepsilon}\,\ln(b/a)$$

$$\textbf{(d)}\quad R = \frac{1}{2\pi\sigma\delta}\left(\frac{1}{a} + \frac{1}{b}\right),\quad L = \frac{\mu}{2\pi}\ln(b/a) + \frac{\mu_c\delta}{4\pi}\left(\frac{1}{a} + \frac{1}{b}\right)$$

Flow: get the TEM fields from Lec 10 → integrate the Poynting vector for P → integrate the wall loss for γ and R → integrate E and H for Z_0 → integrate the magnetic energy for L. **Every part is one integral over the same two fields.**

Step 0 — The TEM fields in a coax

Lec 10 gives the TEM mode: with $B_z = 0$ **and** $E_z = 0$,

$\vec{E}_t = -\nabla_t\phi,\quad \nabla_t^2\phi = 0,\quad k = \omega\sqrt{\mu\varepsilon}$ (Lec 10, s13–15)

So it is a **2D electrostatics problem** in the cross-section. For two concentric circles the solution depends on ρ only:

$$\phi(\rho) = A\ln\rho + B \;\Rightarrow\; E_\rho = -\frac{d\phi}{d\rho} = -\frac{A}{\rho}$$ *everything ∝ 1/ρ*

For a TEM wave **H** is azimuthal and tied to **E** by the wave impedance $Z = \sqrt{\mu/\varepsilon}$. Writing H_0 for its value at $\rho = a$:

$$H_\phi = H_0\,\frac{a}{\rho},\qquad E_\rho = \sqrt{\mu/\varepsilon}\,H_0\,\frac{a}{\rho}$$

These two lines are the whole of Step 0 — every part of the problem is an integral of them. The 1/ρ is forced by $\nabla_t^2\phi = 0$ in cylindrical symmetry; it is the same 1/ρ as the field of a line charge.

PART (a) — time-averaged power flow along the line

Step 1 — The Poynting vector

Power flows along z. The time-averaged Poynting vector for complex amplitudes is

$$\langle \vec{S} \rangle = \frac{1}{2}\,\mathrm{Re}\left(\vec{E} \times \vec{H}^{*}\right)$$ Jackson §6.9

With **E** radial and **H** azimuthal, $\hat{\boldsymbol{\rho}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}$, so the flow is purely along the line:

$$\langle S_z \rangle = \frac{1}{2} E_\rho H_\phi^{*} = \frac{1}{2}\sqrt{\mu/\varepsilon}\,|H_0|^2\,\frac{a^2}{\rho^2}$$

Step 2 — Integrate over the cross-section

The dielectric fills the annulus $a \le \rho \le b$. A ring of radius ρ has area $2\pi\rho\,d\rho$:

$$P = \int_a^b \frac{1}{2}\sqrt{\mu/\varepsilon}\,|H_0|^2\,\frac{a^2}{\rho^2}\,2\pi\rho\,d\rho$$

The ρ^2 downstairs cancels one ρ upstairs, leaving $d\rho/\rho$:

$$= \pi\sqrt{\mu/\varepsilon}\,|H_0|^2 a^2 \int_a^b \frac{d\rho}{\rho} = \pi\sqrt{\mu/\varepsilon}\,|H_0|^2 a^2\,\ln\!\left(\frac{b}{a}\right)$$

That is where the ln(b/a) comes from *— it is just ∫dρ/ρ. It appears again in (c) and (d) for the same reason.*

ANSWER — PART (a)

$$P = \sqrt{\mu/\varepsilon}\,\pi a^2 |H_0|^2 \ln\left(\frac{b}{a}\right)$$

PART (b) — attenuation along the line

Step 1 — Why the power decays at all

The dielectric is lossless, so the only loss is **ohmic heating in the metal walls**. The wall is not a perfect conductor, so the surface field drives a current in a skin layer of depth δ.

$$\frac{dP_{loss}}{dA} = \frac{1}{2\sigma\delta}\,|H_{\parallel}|^2$$ *power dissipated per unit wall area* (Jackson §8.1)

Step 2 — Add up both walls

The tangential field at each wall is H_ϕ evaluated there, and each wall contributes its own circumference per unit length:

wall	$\lvert H_{\parallel}\rvert$	circumference	loss per unit length
inner, $\rho = a$	H_o	$2\pi a$	$\pi a H_o^2/\sigma\delta$
outer, $\rho = b$	$H_o a/b$	$2\pi b$	$\pi a^2 H_o^2/\sigma\delta b$

$$-\frac{dP}{dz} = \frac{\pi a^2 H_o^2}{\sigma\delta}\left(\frac{1}{a} + \frac{1}{b}\right)$$ *a^2 factored out of both rows*

Step 3 — Convert a loss rate into an exponent

If $P(z) = P_o e^{-2\gamma z}$, then differentiating gives $-dP/dz = 2\gamma P$. So

$$\gamma = \frac{-dP/dz}{2P}$$

Divide the Step 2 result by twice the part (a) answer. The $\pi a^2 H_o^2$ cancels completely:

$$\gamma = \frac{\pi a^2 H_o^2 (1/a + 1/b)/\sigma\delta}{2\sqrt{\mu/\varepsilon}\,\pi a^2 H_o^2 \ln(b/a)}$$

ANSWER — PART (b)

$$\gamma = \frac{1}{2\sigma\delta}\sqrt{\varepsilon/\mu}\,\frac{1/a + 1/b}{\ln(b/a)}$$

$1/\sqrt{\mu/\varepsilon}$ was written as $\sqrt{\varepsilon/\mu}$.

PART (c) — characteristic impedance $Z_0 = V/I$

The problem **defines** Z_0 as the voltage between the cylinders divided by the axial current in one of them. So compute each separately from the Step 0 fields.

Step 1 — The voltage

Integrate the radial field across the gap:

$$V = \int_a^b E_\rho \, d\rho = \sqrt{\mu/\varepsilon}\, H_0 a \int_a^b \frac{d\rho}{\rho} = \sqrt{\mu/\varepsilon}\, H_0 a \ln(b/a)$$

Step 2 — The current

Ampère's law round the inner conductor — H_ϕ is constant on that circle:

$$I = \oint \vec{H}\cdot d\vec{l} = H_0 \cdot 2\pi a$$

Step 3 — Divide

$$Z_0 = \frac{V}{I} = \frac{\sqrt{\mu/\varepsilon}\, H_0 a \ln(b/a)}{2\pi a H_0}$$

H_0 and a both cancel

ANSWER — PART (c)

$$Z_0 = \frac{1}{2\pi}\sqrt{\mu/\varepsilon}\,\ln\left(\frac{b}{a}\right)$$

Depends only on the **geometry ratio** b/a and the dielectric — not on a and b separately, and not on frequency. That is why coax is sold as "50 Ω" regardless of cable thickness.

PART (d) — series resistance and inductance per unit length

Step 1 — R from the wall loss

A resistance R per unit length carrying peak current I dissipates $\frac{1}{2} R|I|^2$ per unit length.

Set that equal to the wall loss already computed in part (b):

$$\frac{1}{2} R|I|^2 = \frac{\pi a^2 H_o^2}{\sigma\delta}\left(\frac{1}{a} + \frac{1}{b}\right), \qquad |I|^2 = (2\pi a H_o)^2 = 4\pi^2 a^2 H_o^2$$

Solve for R — $a^2 H_o^2$ cancels:

$$R = \frac{2\pi a^2 H_o^2 (1/a + 1/b)/\sigma\delta}{4\pi^2 a^2 H_o^2} = \frac{1}{2\pi\sigma\delta}\left(\frac{1}{a} + \frac{1}{b}\right)$$

*This is just "resistance = resistivity × length / area" for a conductor whose effective thickness is the skin depth δ: each wall carries current in a strip of width 2πa (or 2πb) and depth δ, and the two are in **series**, so their resistances add.*

Step 2 — L from the stored magnetic energy

An inductance L per unit length stores $\frac{1}{4}L|I|^2$ on time average. The energy sits in **two** places:

(i) In the dielectric, between the conductors:

$$W_{ext} = \frac{\mu}{4}\int_a^b |H_\phi|^2\, 2\pi\rho\, d\rho = \frac{\pi\mu a^2 H_0^2}{2}\ln(b/a)$$ *same $\int d\rho/\rho$ as part (a)*

(ii) Inside the metal, in the skin layer. The field decays as $H_s e^{-x/\delta}$ going into the wall, so per unit wall area:

$$\frac{\mu_c}{4}\int_0^\infty H_s^2 e^{-2x/\delta}\, dx = \frac{\mu_c \delta H_s^2}{8}$$ *$\int e^{-2x/\delta}dx = \delta/2$*

Add both walls, as in part (b):

$$W_{int} = \frac{\pi\mu_c\delta a^2 H_0^2}{4}\left(\frac{1}{a}+\frac{1}{b}\right)$$

Set $\frac{1}{4}L|I|^2 = W_{ext} + W_{int}$ and divide by $\pi^2 a^2 H_0^2$:

ANSWER — PART (d)

$$R = \frac{1}{2\pi\sigma\delta}\left(\frac{1}{a}+\frac{1}{b}\right)$$

$$L = \frac{\mu}{2\pi}\ln\left(\frac{b}{a}\right) + \frac{\mu_c\delta}{4\pi}\left(\frac{1}{a}+\frac{1}{b}\right)$$

*First term = flux **between** the conductors. Second = flux that leaks a depth δ **into** them — exactly the correction the question mentions.*

■ END OF JACKSON 8.2 ■

Jackson 8.4 — Hollow circular waveguide

> TE and TM waves propagate along a hollow circular cylinder of inner radius R and conductivity σ. **(a)** Find the cutoff frequencies of the TE and TM modes; determine the lowest (dominant) one and the ratios of the next four higher modes to it, assuming $\sigma \to \infty$. **(b)** Calculate the attenuation constants for the lowest two distinct modes and plot them against frequency.

ANSWER

(a) $\omega_c = \dfrac{x}{R\sqrt{\mu\varepsilon}}$, $x = x_{mn}$ (TM: J_m zero) or x'_{mn} (TE: J'_m zero)

dominant = **TE$_{11}$**, $x'_{11} = 1.841 \rightarrow \omega_c = 1.841/(R\sqrt{\mu\varepsilon})$

next four: TM$_{01}$ **1.306**, TE$_{21}$ **1.659**, TE$_{01}$ and TM$_{11}$ **2.081** (degenerate)

(b) $\alpha_{TM} = \dfrac{R_s}{R\eta}\,\dfrac{1}{\sqrt{1-(\omega_c/\omega)^2}}$

$\alpha_{TE} = \dfrac{R_s}{R\eta}\,\dfrac{1}{\sqrt{1-(\omega_c/\omega)^2}}\left[\left(\dfrac{\omega_c}{\omega}\right)^2 + \dfrac{m^2}{x'^2 - m^2}\right]$

$R_s = 1/\sigma\delta = \sqrt{\mu_c\omega/2\sigma}$ (surface resistance), $\eta = \sqrt{\mu/\varepsilon}$.

Flow: the lecture already gives the 2D equation and the wall condition for each mode. In a **circle** those become Bessel equations → the wall condition picks out Bessel zeros → each zero is one mode with its own cutoff.

PART (a) — cutoff frequencies

Step 1 — The two mode equations, from lecture

Lec 10 has already reduced each mode to a 2D problem in the cross-section:

TM ($B_z = 0$): $\nabla_t^2 E_z + \gamma^2 E_z = 0$, $E_z = 0$ on the wall (Lec 10, s17)

TE ($E_z = 0$): $\nabla_t^2 B_z + \gamma^2 B_z = 0$, $\partial B_z/\partial n = 0$ on the wall (Lec 10, s18)

$\gamma^2 = \mu\varepsilon\omega^2 - k^2$ *the same γ in both* (Lec 10, s17)

*The **only** difference between TE and TM is the wall condition: the function itself vanishes for TM, its normal derivative vanishes for TE. Everything else below is identical.*

Step 2 — A circle forces cylindrical coordinates

Write ψ for whichever of E_z or B_z applies. In polar coordinates

$$\nabla_t^2 = \frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial}{\partial\rho}\right) + \frac{1}{\rho^2}\frac{\partial^2}{\partial\phi^2}$$

Separate, $\psi = f(\rho)e^{im\phi}$. **m must be an integer**, otherwise ψ would not come back to itself after going once round the circle:

$$f'' + \frac{1}{\rho}f' + \left(\gamma^2 - \frac{m^2}{\rho^2}\right)f = 0 \quad \textit{Bessel's equation}$$

Its two solutions are $J_m(\gamma\rho)$ and $Y_m(\gamma\rho)$. The guide is **hollow with no inner conductor**, so $\rho = 0$ is inside the region — and Y_m blows up there. Keep only

$$\psi = J_m(\gamma\rho)\,e^{im\phi}$$

*This is the one place Bessel functions are unavoidable — they are simply "sine and cosine for a circle". In a **rectangular** guide the same separation gives ordinary sines, which is the case done in* (Lec 9, s17).

Step 3 — The wall condition picks out the zeros

TM: $\psi(R) = 0 \Rightarrow J_m(\gamma R) = 0 \Rightarrow \gamma R = x_{mn}$ *n^{th} zero of J_m*

TE: $\partial\psi/\partial\rho\big|_R = 0 \Rightarrow J'_m(\gamma R) = 0 \Rightarrow \gamma R = x'_{mn}$ *n^{th} zero of J'_m*

Each zero is **one mode**. That is what m and n label.

Step 4 — From γ to the cutoff frequency

The wave propagates only if k is **real**, i.e. $k^2 > 0$:

$$k^2 = \mu\varepsilon\omega^2 - \gamma^2 > 0 \quad \Rightarrow \quad \omega > \frac{\gamma}{\sqrt{\mu\varepsilon}}$$

Below that, k is imaginary and the field dies away instead of travelling. The changeover is the **cutoff**:

$$\omega_c = \frac{\gamma}{\sqrt{\mu\varepsilon}} = \frac{x}{R\sqrt{\mu\varepsilon}} \qquad x = x_{mn} \text{ for TM, } x'_{mn} \text{ for TE}$$

Smaller x → lower cutoff. *So ranking the modes just means ranking the Bessel zeros.*

Step 5 — Rank the zeros

mode	$x = \gamma R$	ω_c / $\omega_c(TE_{11})$
TE_{11}	**1.8412**	**1.000** ← dominant
TM_{01}	2.4048	1.306
TE_{21}	3.0542	1.659
TE_{01}	3.8317	2.081
TM_{11}	3.8317	2.081
TE_{31}	4.2012	2.282

ANSWER — PART (a)

dominant: **TE_{11}**, $\omega_c = \dfrac{1.841}{R\sqrt{\mu\varepsilon}}$

next four: **1.306** (TM_{01}), **1.659** (TE_{21}), **2.081** (TE_{01} and TM_{11})

TE_{01} and TM_{11} are ***degenerate*** — a coincidence of Bessel functions, $J'_0 = -J_1$, so the zeros of J'_0 and J_1 are the same numbers. For R = 1 cm in vacuum, the dominant cutoff is ***8.785 GHz.***

PART (b) — attenuation of the lowest two modes

The two lowest **distinct** modes are $\mathbf{TE_{11}}$ ($x' = 1.841$) and $\mathbf{TM_{01}}$ ($x = 2.405$), from the table above.

Step 1 — Same method as 8.2(b)

Wall loss over power flow — identical logic, only the fields differ:

$$\alpha = \frac{-dP/dz}{2P}, \qquad -\frac{dP}{dz} = \frac{R_s}{2}\oint |H_\parallel|^2\, dl, \qquad R_s = \frac{1}{\sigma\delta} = \sqrt{\mu_c \omega / 2\sigma}$$ Jackson §8.1

Carrying out both integrals with the Bessel fields gives:

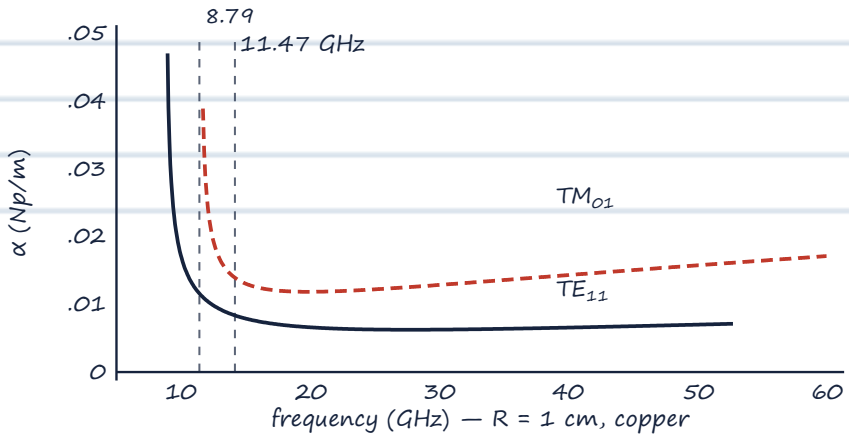
$$\alpha_{TM} = \frac{R_s}{R\eta}\,\frac{1}{\sqrt{1-(\omega_c/\omega)^2}}$$

$$\alpha_{TE} = \frac{R_s}{R\eta}\,\frac{1}{\sqrt{1-(\omega_c/\omega)^2}}\left[\left(\frac{\omega_c}{\omega}\right)^2 + \frac{m^2}{x'^2 - m^2}\right]$$

For our two modes: TM_{01} has $m = 0$, so no bracket. TE_{11} has $m = 1$ and $x' = 1.841$, so the extra term is $1/(1.841^2-1) = \mathbf{0.419}$.

Step 2 — Read the shape

1. Both **diverge at cutoff** — $1/\sqrt{1-(\omega_c/\omega)^2} \to \infty$ as $\omega \to \omega_c$.
2. Far above cutoff $R_s \propto \sqrt{\omega}$, so both **rise again**.
3. So each has a **minimum** in between — the best operating frequency.



Dashed lines are the two cutoffs. Each curve rises steeply at its own cutoff, dips, then climbs as $\sqrt{\omega}$.

ANSWER — PART (b)

TE_{11}: minimum $\alpha = 0.0063$ Np/m at $\omega/\omega_c = 3.16$ (27.7 GHz)

TM_{01}: minimum $\alpha = 0.0119$ Np/m at $\omega/\omega_c = 1.73$ (19.8 GHz)

TE_{11} is the lower-loss mode over the whole band — which, with its lower cutoff, is why circular guides are run in TE_{11}.

■ END OF JACKSON 8.4 ■